

KL Divergence based Retrieval Model

Introduction to Information Retrieval

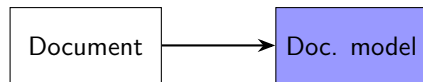
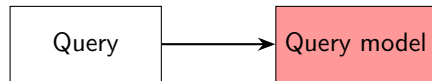
Dwaipayan Roy

Department of Computational and Data Sciences



Language Modeling for IR

Modeling Documents and Queries



Three Probabilistic Perspectives

- **(a) Query Likelihood:** Probability that the query was generated by the document model
- **(b) Document Likelihood:** Probability that the document was generated by the query model
- **(c) Model Comparison:** Direct comparison between query and document models

KL Divergence: Measuring Distribution Similarity

The Fundamental Problem

Given two probability distributions $P(x)$ and $Q(x)$ defined on the same probability space $\mathcal{X}$, how do we measure their *similarity* or *dissimilarity*?

Kullback-Leibler Divergence

$$KLD(P||Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}$$

Key Properties

- **Non-negative:** $KLD(P||Q) \geq 0$
- **Identity:** $KLD(P||Q) = 0$ iff $P = Q$
- **Asymmetric:** $KLD(P||Q) \neq KLD(Q||P)$
- **Not a metric:** No triangle inequality

Interpretation

- Measures information loss when using Q to approximate P
- Expected log-likelihood ratio
- Inverse of similarity measure

KL Divergence for Retrieval

Language Model Perspective

- M_Q, M_D : unigram language model associated with the query Q and the document D respectively.
- $p(Q|M_Q), p(D|M_D)$: generative model that generates query Q and document D respectively.
- Relevance: How similar are the query and document models?

The Retrieval Scoring Function

$$KLD(M_Q||M_D) = \sum_w p(w|M_Q) \log \frac{p(w|M_Q)}{p(w|M_D)}$$

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KL Divergence for Retrieval

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- occurs in document D.

Complete KL-Divergence Retrieval Algorithm

Step-by-Step Process

- 1 **Preprocessing:** Build collection language model $p(w|C)$
- 2 **Document Modeling:** For each document D , estimate smoothed model:

$$p(w|M_D) = \frac{tf(w, D) + \mu p(w|C)}{|D| + \mu}$$

- 3 **Query Modeling:** Estimate smoothed query model:

$$p(w|M_Q) = \lambda \frac{tf(w, Q)}{|Q|} + (1 - \lambda)p(w|C)$$

- 4 **Scoring:** Compute KL-divergence score:

$$score(Q, D) = \sum_w p(w|M_Q) \log p(w|M_D)$$

- 5 **Ranking:** Sort documents by decreasing score

Jensen-Shannon Divergence: Symmetric Alternative

Motivation

- KL-divergence is asymmetric: $KLD(P||Q) \neq KLD(Q||P)$.
- Need symmetric measure for document-document similarity.

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Jensen-Shannon Divergence Definition

$$JSD(P||Q) = \frac{1}{2}KLD(P||\theta_M) + \frac{1}{2}KLD(Q||\theta_M)$$

where $\theta_M = \frac{1}{2}(P + Q)$ is the mixture distribution

Properties

- **Symmetric:** $JSD(P||Q) = JSD(Q||P)$
- **Bounded:** $0 \leq JSD(P||Q) \leq 1$

Jensen-Shannon Divergence

between a pair of document

- Jensen-Shannon Divergence between D_1 and D_2 :

$$JSD(D_1||D_2) = \frac{1}{2}KLD(D_1||\theta_M) + \frac{1}{2}KLD(D_2||\theta_M)$$

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- $JSD(D_1||D_2) == JSD(D_2||D_1)$

Applications

- Use **KL-Divergence**: when the entity being compared is of **dissimilar length**.
 - ▶ e.g.: compute **document-query** similarity.
- Use **JS-Divergence**: when the entity being compared is of **similar length**.
 - ▶ e.g.: compute **document-document** similarity.